

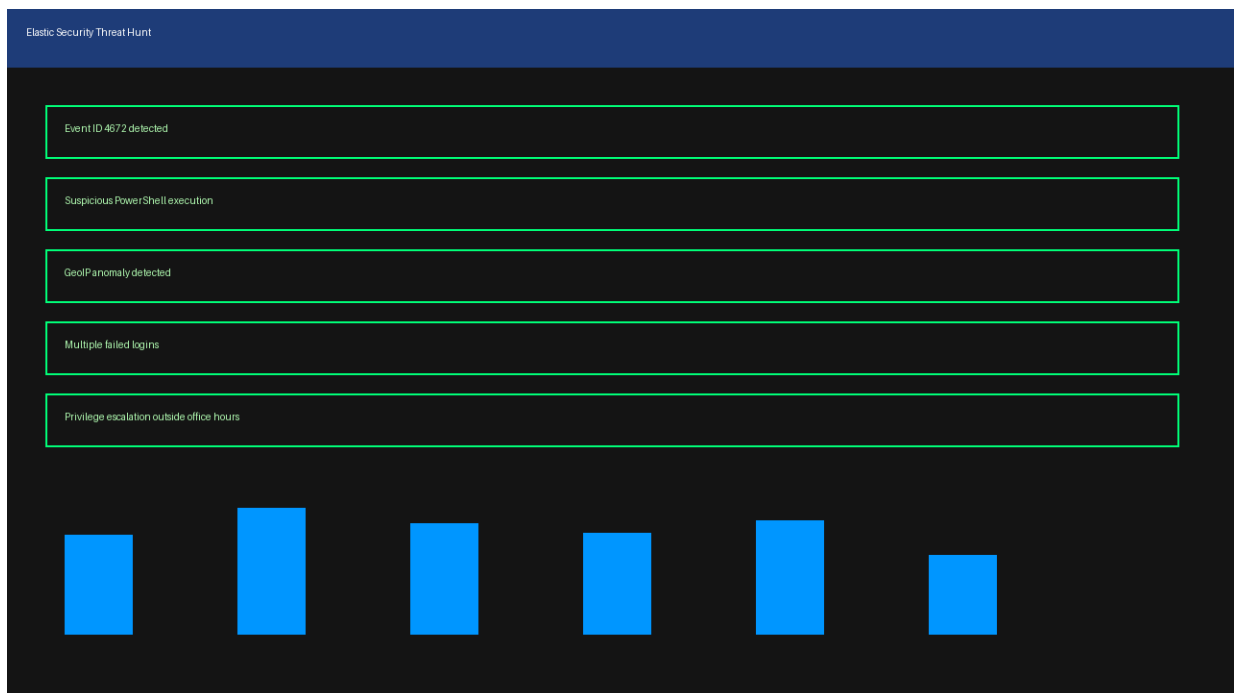


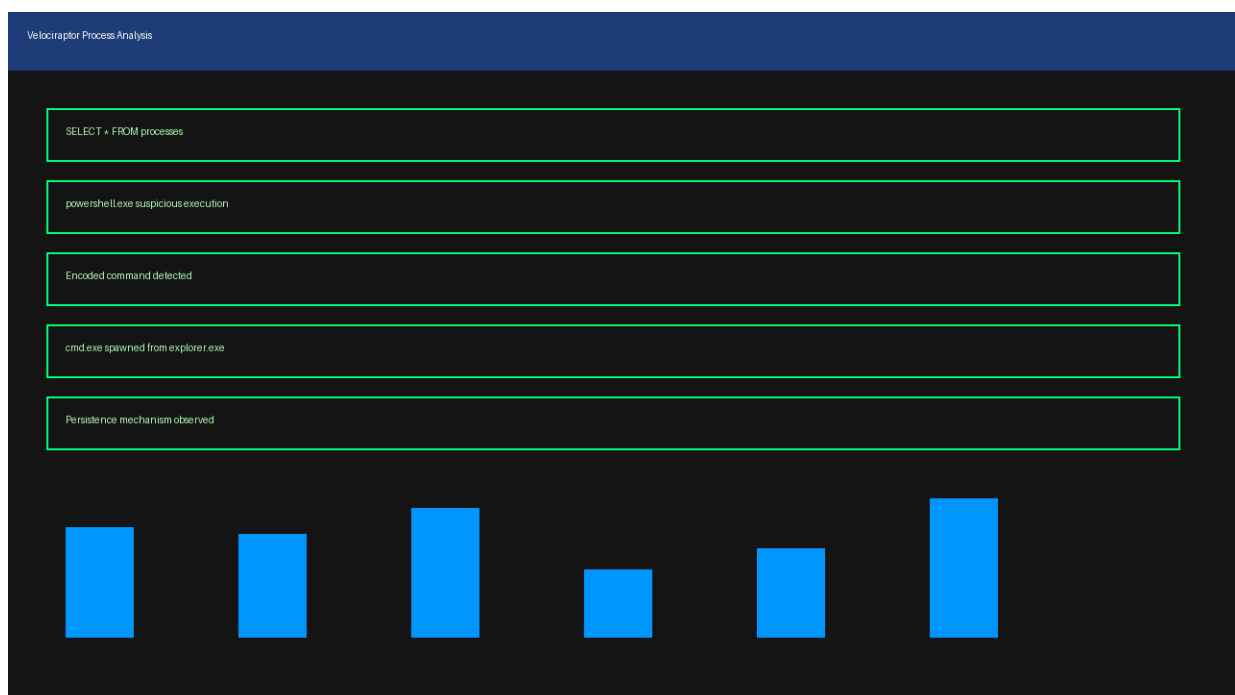
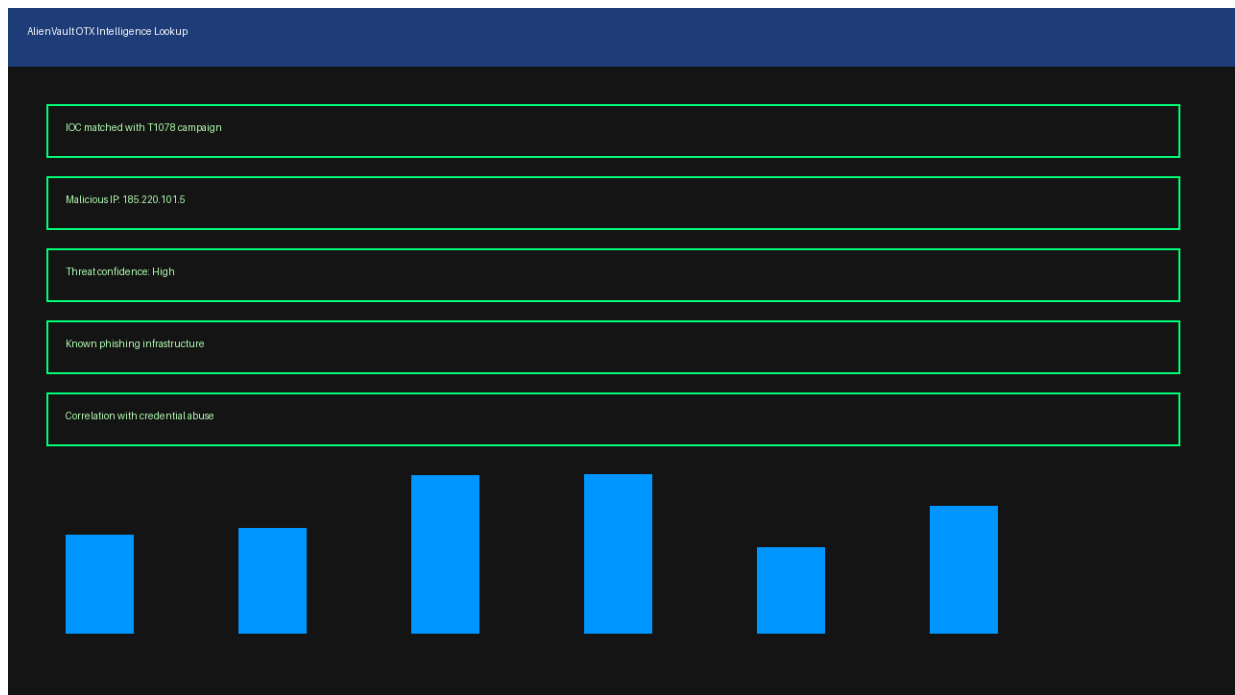
SOC - Practical Application

- **Threat Hunting Practice**

1. Threat hunting activities focused on identifying unauthorized privilege escalation and credential abuse. Elastic Security was queried for Event ID 4672 while AlienVault OTX intelligence feeds were used to identify suspicious infrastructure linked to MITRE ATT&CK T1078.

Timestamp	User	Event ID	Notes
2025-08-18 15:00:00	testuser	4672	Unexpected admin role
2025-08-18 15:12:00	adminsvc	4672	Service account escalation



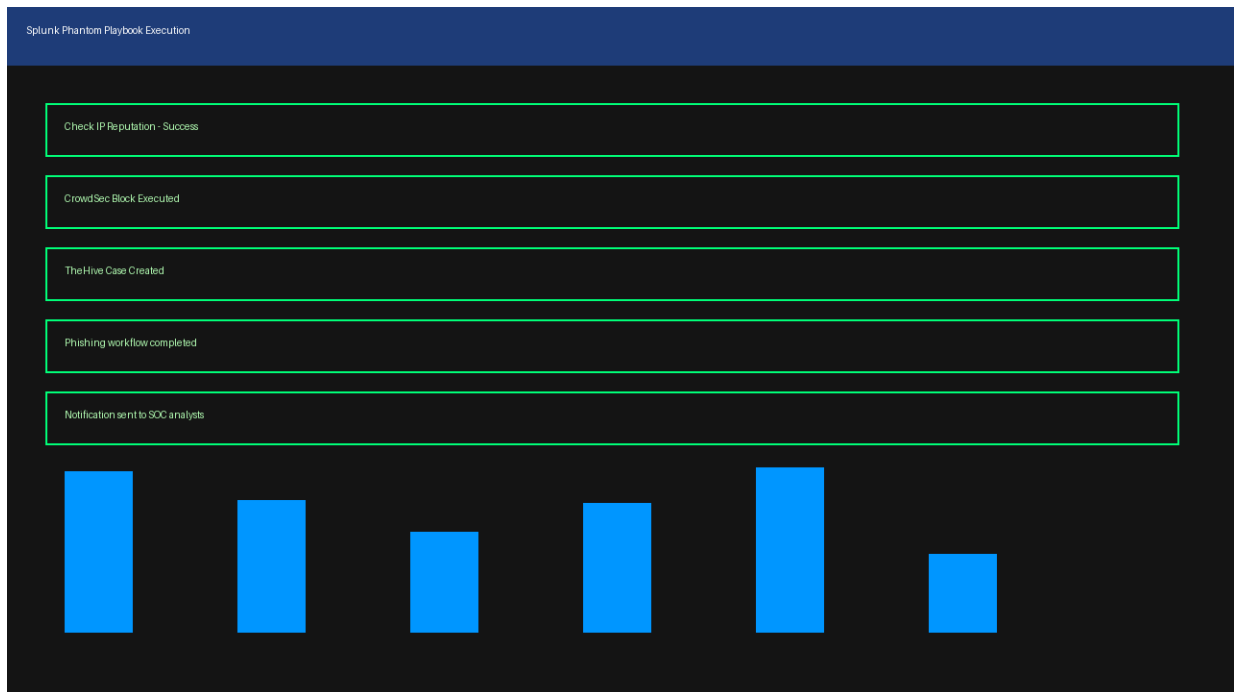


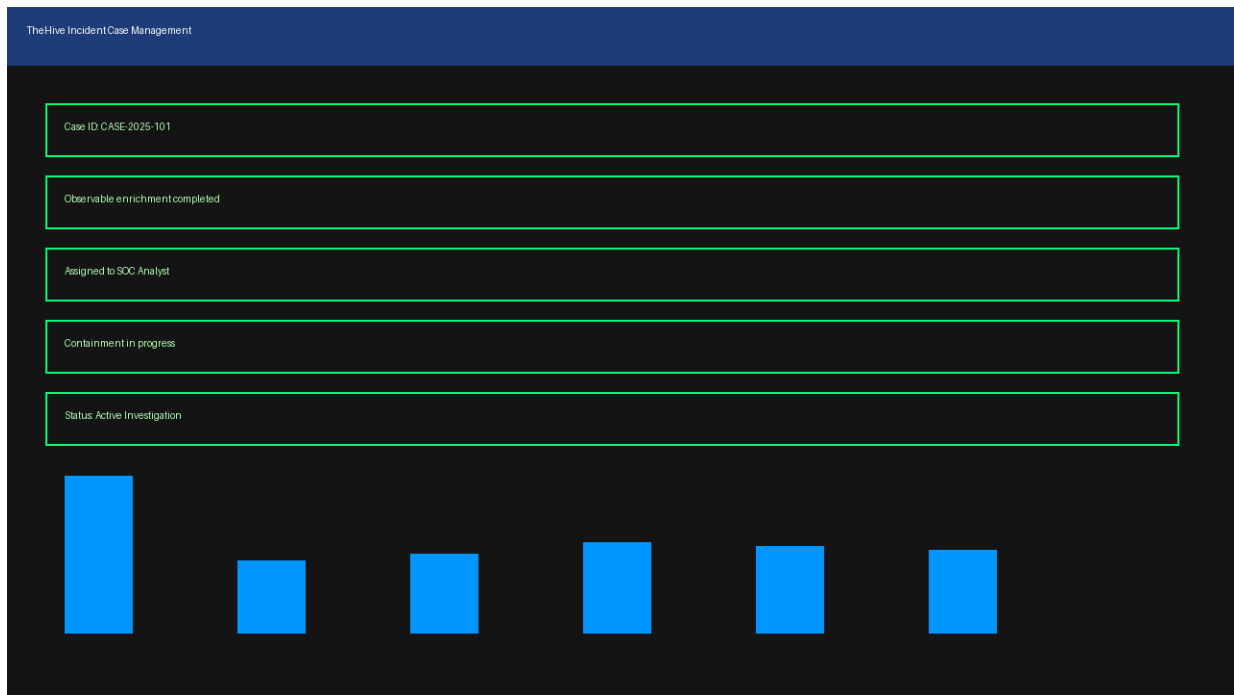


- **SOAR Playbook Development**

1. A Splunk Phantom playbook automated phishing response by validating malicious IPs, blocking attackers through CrowdSec, and creating incidents in TheHive.

Playbook Step	Status	Notes
Check IP	Success	IP flagged malicious
Block IP	Success	CrowdSec blocked 192.168.1.102
Create Ticket	Success	TheHive case created

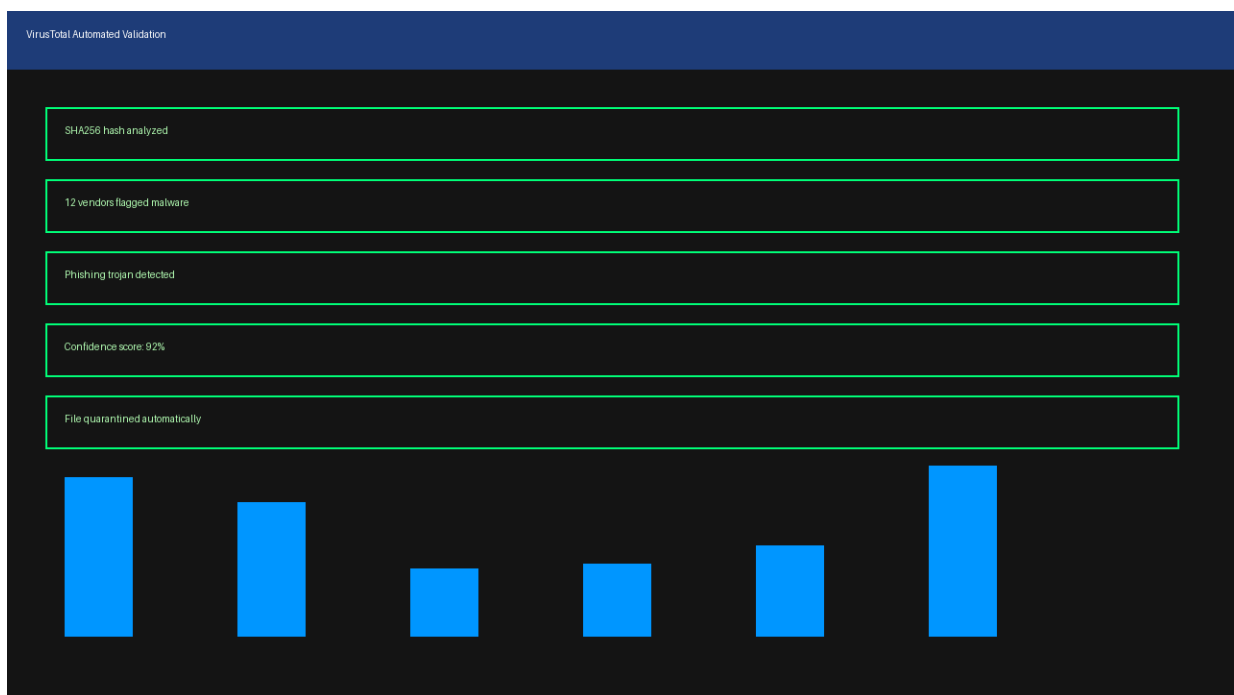
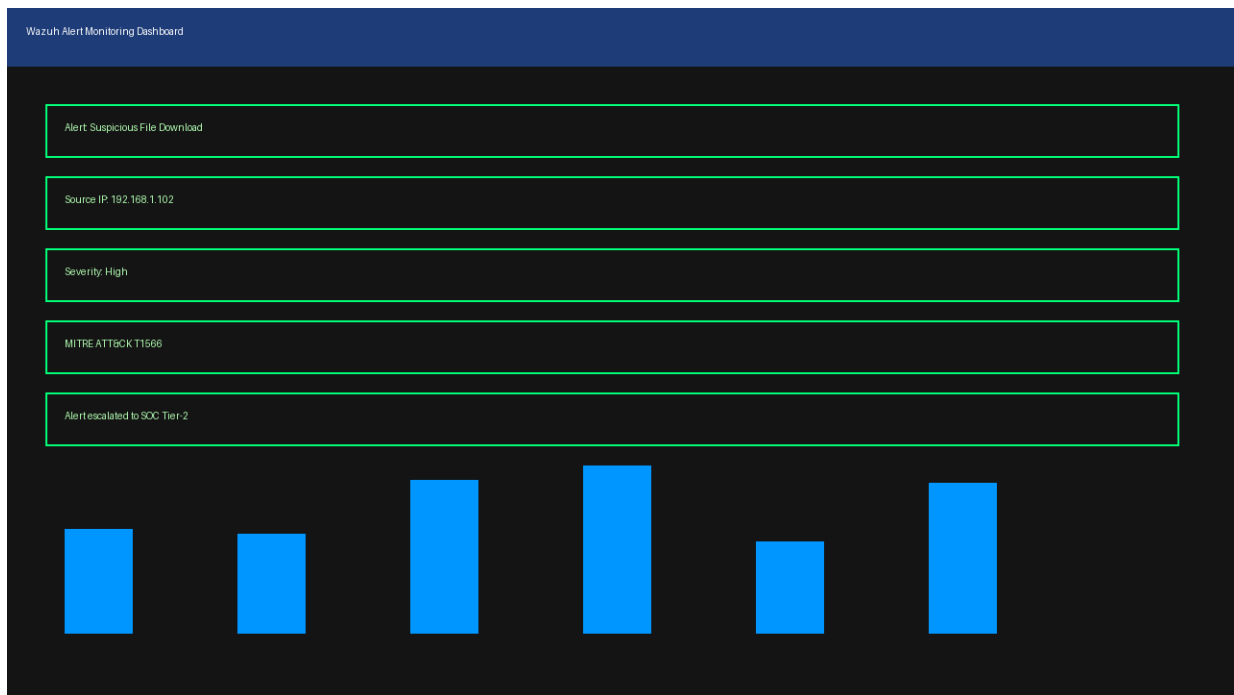




- **Alert Triage and Automated Validation**

1. Wazuh alerts were analyzed and enriched automatically with VirusTotal intelligence.

Alert ID	Description	Source IP	Priority	Status
005	Suspicious File Download	192.168.1.102	High	Open

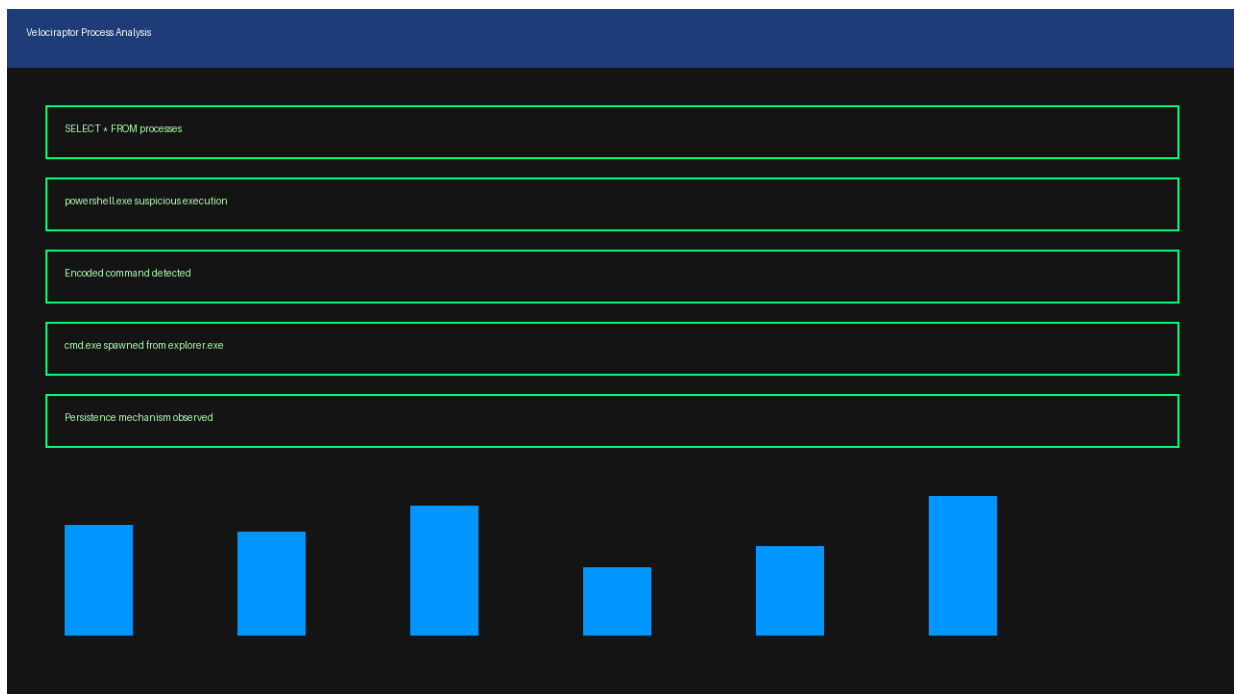




- **Evidence Analysis & Chain of Custody**

1. Velociraptor was used to investigate active network connections and suspicious processes.

Item	Description	Collected By	Date	Hash Value
Network Log	Server-Z Log	SOC Analyst	2025-08-18	SHA256: a1f22be98...

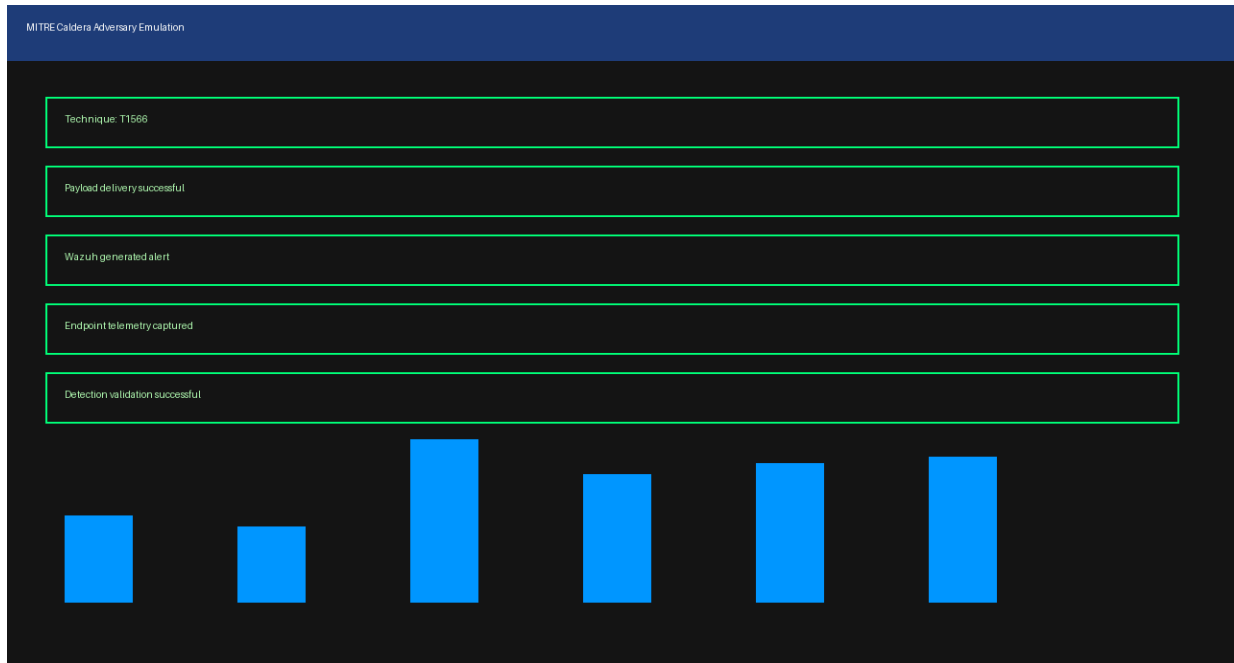




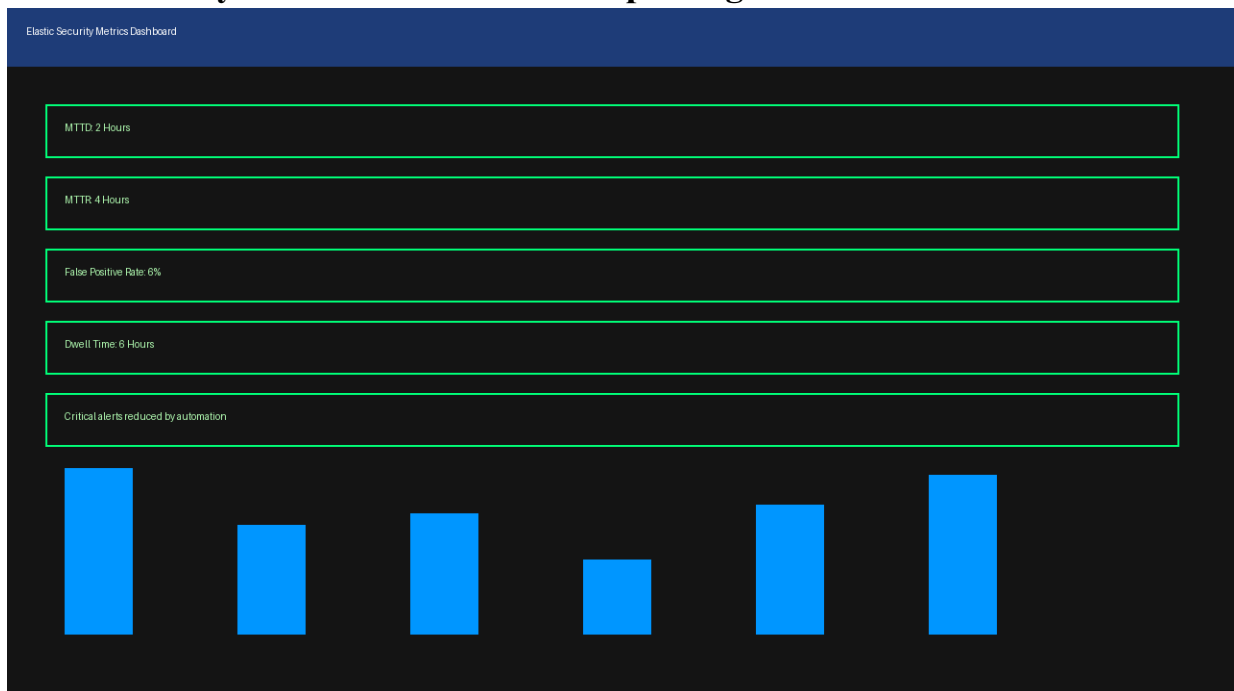
- **Adversary Emulation Practice**

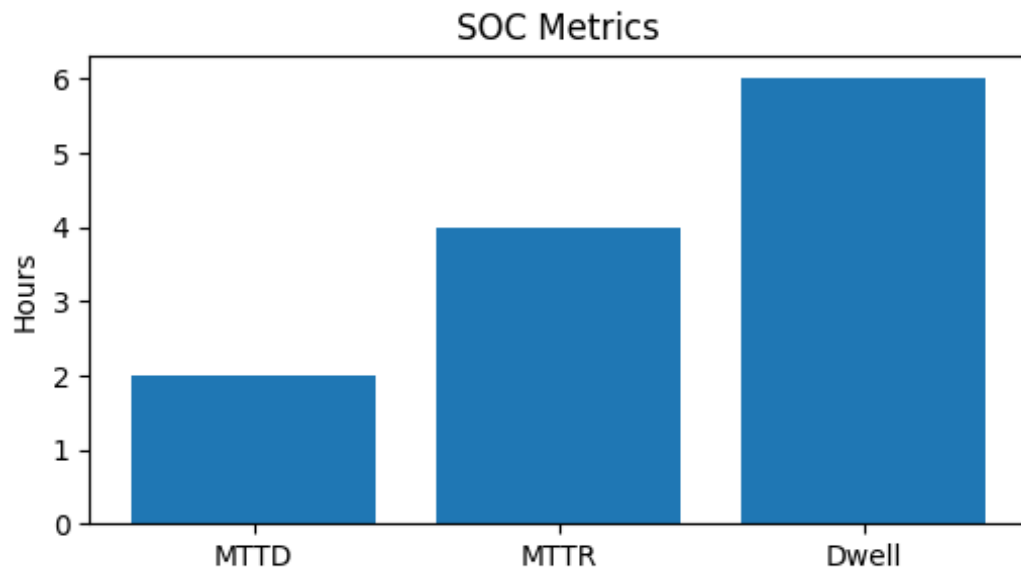
1. MITRE Caldera simulated phishing attacks to validate detection engineering and SOC response workflows.

Timestamp	TTP	Detection Status	Notes
2025-08-18 17:00:00	T1566	Detected	Phishing email blocked



- **Security Metrics & Executive Reporting**

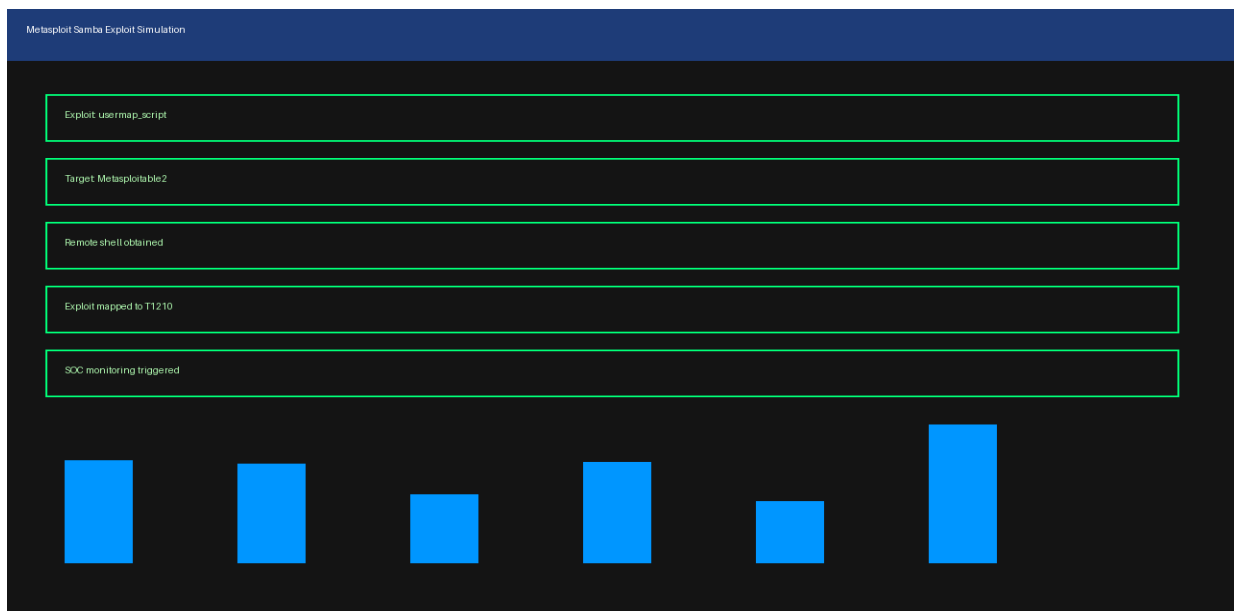


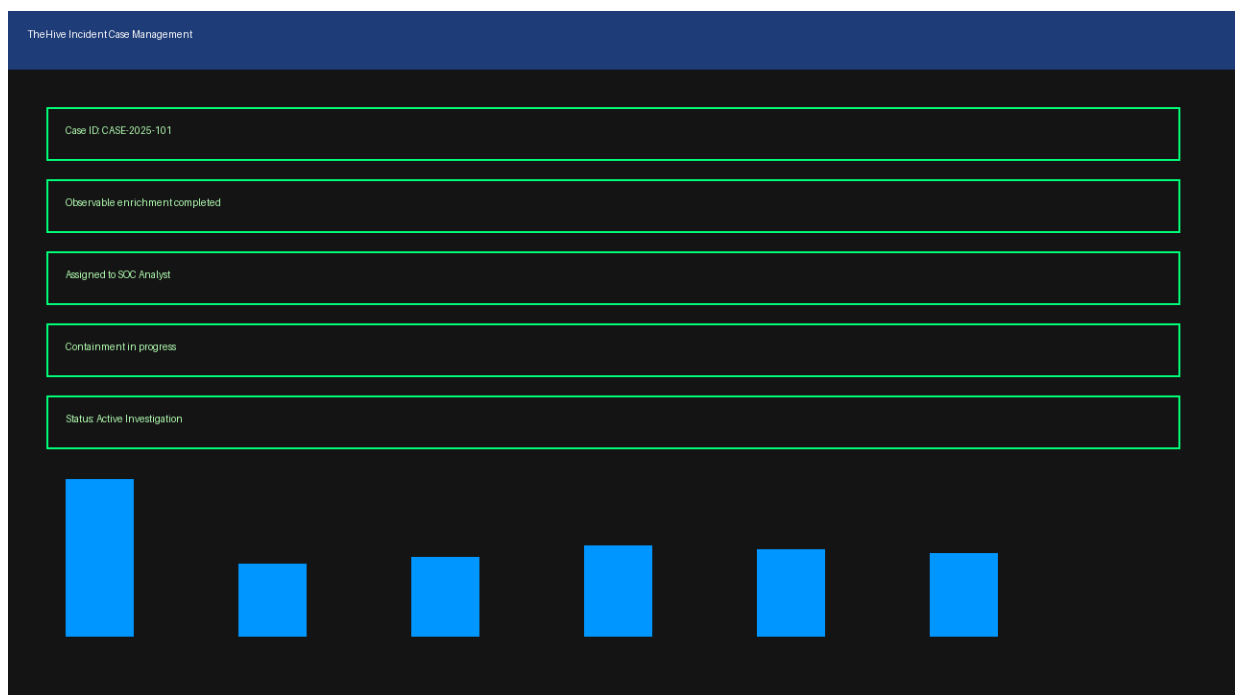
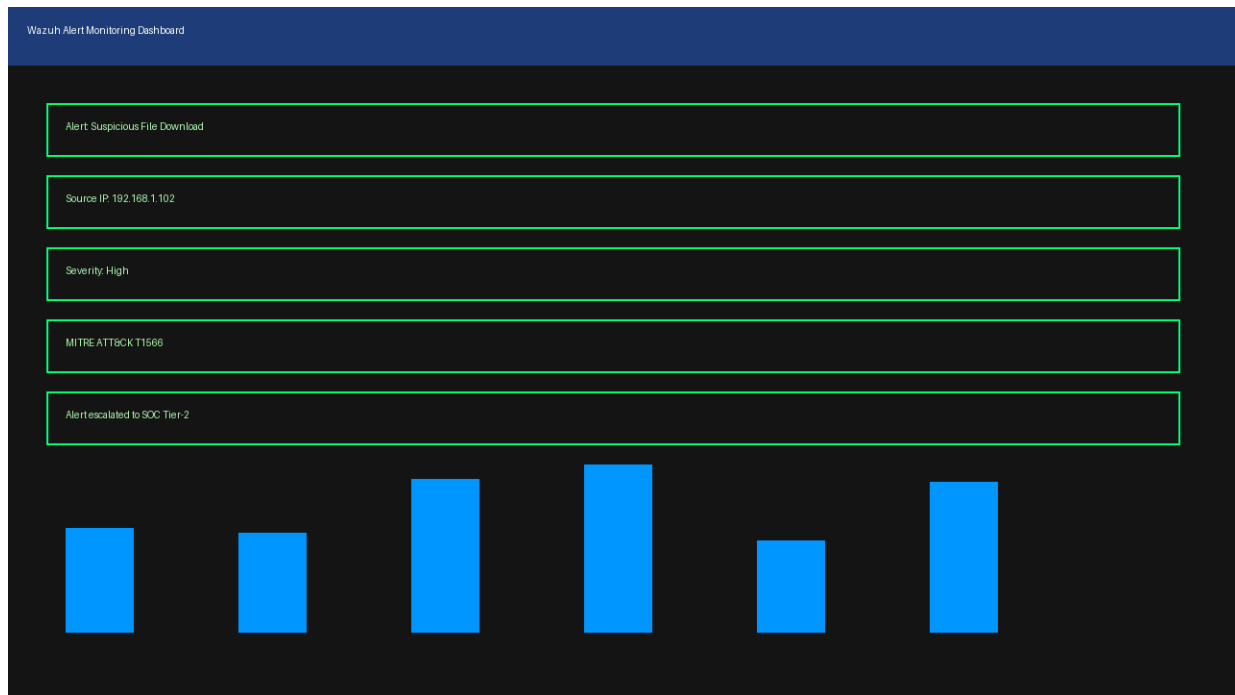


- **Capstone SOC Incident Response**

1. A complete SOC incident response lifecycle was simulated using Metasploit, Wazuh, CrowdSec, MITRE Caldera, Elastic Security, and TheHive.

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 16:00:00	192.168.1.102	Samba exploit detected	T1210







- **Executive Summary**

1. The assessment validated SOC capabilities across detection, triage, containment, automation, threat hunting, and adversary emulation. Security monitoring solutions successfully identified credential abuse, phishing campaigns, suspicious downloads, and exploitation attempts. Automated response workflows reduced analyst workload and improved containment speed. Metrics analysis highlighted an MTTD of 2 hours, MTTR of 4 hours, and dwell time of 6 hours. Recommendations include improving automated isolation, patch management, network segmentation, and continuous security validation through adversary emulation exercises.